

3ART13 LED Fabrication Document

Layer Stack Legend

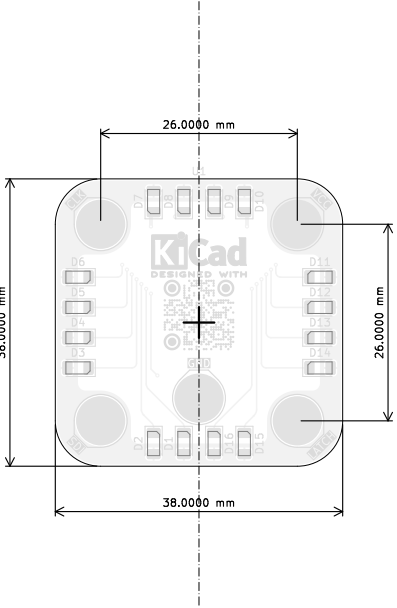
	Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste				Paste Mask	
	F.Silkscreen			Direct Printing	Legend	GBR
	F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	Copper	T.Cu (Sig, PWR)	0.035mm (1.00oz)		Signal	GBR
	Core		1.5mm	FR4_7628	Dielectric	
	Copper	B.Cu (Sig, PWR)	0.035mm (1.00oz)		Signal	GBR
	B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen			Direct Printing	Legend	GBR
	B.Paste				Paste Mask	

Total thickness: 1.61mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - B. Tg 170 C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BAJRAN IMGAGINARY INDUSTRIES.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

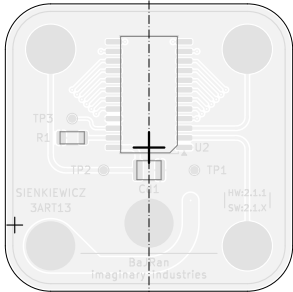
BOARD SIZE	38.000 × 38.000 mm
BOARD THICKNESS	1.610 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	N/A mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: c7cddf3
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 7

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Bottom Fabrication (Scale 1:1)



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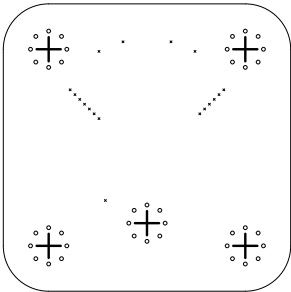
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		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 7

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	18	0,25mm (9,84mils)	PTH	Round	T,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
O	40	0,50mm (19,69mils)	PTH	Round	T,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
+	5	3,20mm (125,98mils)	PTH	Round	T,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
Total 63						

Drill Drawing L1 - L2 (Scale 1:1)



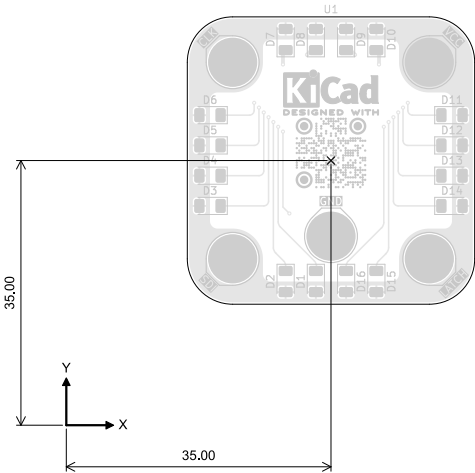
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		Board Name: 3ART13 LED		Project Name: Sienkiewicz		
	Sheet Title: Drill Drawing (L1 - L2)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1+ (Unreleased)	
	Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 7	

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Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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Ref.	Net	X [mm]	Y [mm]
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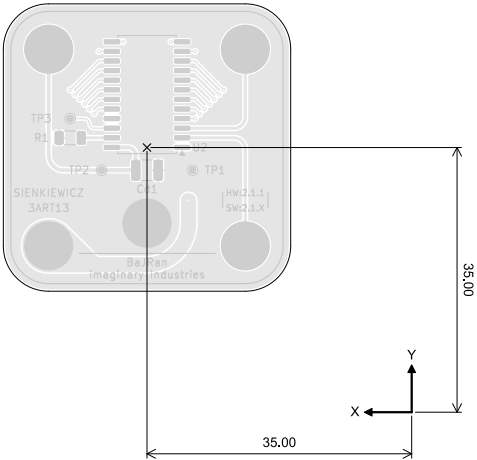
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	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: c7cddf3
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 7

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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	GND	78.50	13.00
TP2	+5V	90.50	13.00
TP3	Net-(U2-SDO)	94.66	19.82

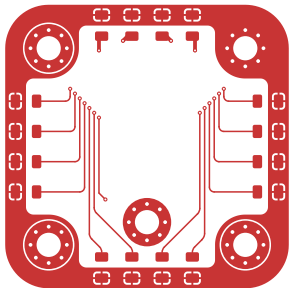


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		Board Name: 3ART13 LED		Project Name: Sienkiewicz		
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1+ (Unreleased)	
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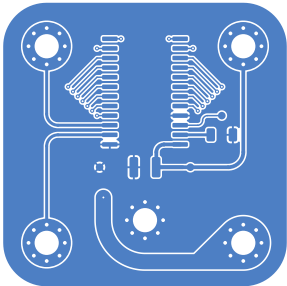
T.Cu (Sig, PWR) (Scale 1:1)



	Comments:	Company: BaJRan		Variant: CHECKED	Git Hash: c7cddf3	
		Board Name: 3ART13 LED		Project Name: Sienkiewicz		
	Sheet Title: T.Cu (Sig, PWR) (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1+ (Unreleased)	
	Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 7	

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B.Cu (Sig, PWR) (Scale 1:1)



	Comments:	Company: BaJRan		Variant: CHECKED	Git Hash: c7cddf3	
		Board Name: 3ART13 LED		Project Name: Sienkiewicz		
	Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1+ (Unreleased)	
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